PROJECT PLAN

Automatic drink dispenser

Phase 1: Concept, component selection and circuit design

Scope of work: Creating the initial project outline, selecting required components, ordering parts, preparing the preliminary schematic and assigning microcontroller pins.

Phase 2: Firmware development for individual prototype components

Scope of work: Developing routines for specific components and bench testing individual modules (weight readout, stepper motor movement and pump activation).

Phase 3: Mechanical construction and circuit assembly

Scope of work: Designing the case, assembling the linear rail, mounting the load cell onto the carriage, connecting the power supply and installing components inside the enclosure.

Phase 4: Main logic and subroutine integration

Scope of work: Integrating subroutines and testing the system logic.

Phase 5: Prototype startup

Scope of work: Testing carriage limit switch operation, positioning the carriage under the selected dispenser, performing liquid dispensing to a target weight and drafting the initial GUI.

Phase 6: Prototype finalization

Scope of work: Implementing full touchscreen control for multiple bottles/recipes, performing final dispensing calibration and organizing the wiring (cable management).

Phase 7: Technical documentation

Scope of work: Preparation of technical documentation.